

Game State & Level Configuration Guide (config.json)

The **config.json** file dictates the state of a match, functioning as either a custom scenario layout or a save-game state. Parameters that list a **default value** are optional and can be safely skipped. Parameters without a default value are strictly mandatory, skipping them will cause a fatal parsing error and the game will not start. Additionally, violating any of the structural or logical constraints below will abort the loading sequence.

1. Root Structure

The root JSON object defines the global game environment and lists all participants.

Key	Type	Description / Constraints
mapWidth	Integer	Absolute X boundary. Must be at least 5. Defaults to 10.
mapHeight	Integer	Absolute Y boundary. Must be at least 5. Defaults to 10.
currentPlayerName	String	Defines whose turn it is upon loading. Must exactly match a name in the players array. Defaults to the first player in the players array
players	Array	Must contain exactly 2 player objects.
entities	Array	Contains all map objects (Cities, Troops). Overlapping restrictions apply.

Bare minimum config with all default values:

```
"players": [  
  {  
    "name": "Player1"  
  },  
  {  
    "name": "Player2"  
  }  
],  
"entities": [  
  {  
    "entityType": "City",
```

```

        "x": 0,
        "y": 0,
        "ownerName": "Player1"
    },
    {
        "entityType": "City",
        "x": 9,
        "y": 9,
        "ownerName": "Player2"
    }
]
}

```

Full config with custom user's values:

```

{
    "mapWidth": // your int value,
    "mapHeight": // your int value,
    "currentPlayerName": // your String value matching one of the
players names,
    "players": [
        {
            "name": // your String value,
            "balance": // your int value
        },
        {
            "name": // your String value,
            "balance": // your int value
        }
    ],
    "entities": [
        {
            "entityType": "City",
            "x": // your int value,
            "y": // your int value,
            "ownerName": // your String value matching one of the
players names,
            "cityLevel": // "LEVEL_1", "LEVEL_2" or "LEVEL_3"
        },
        {
            "entityType": "City",
            "x": // your int value,
            "y": // your int value,
            "ownerName": // your String value matching that other
player name,
            "cityLevel": // "LEVEL_1", "LEVEL_2" or "LEVEL_3"
        }
    ]
}

```

```
    },  
    // troops config  
  ]  
}
```

2. Player Object Specification

Defines the state of the 2 participants.

Key	Type	Description / Constraints
name	String	Unique identifier for the player. Cannot be empty.
balance	Integer	Financial state. Must be 0 or higher. Defaults to 0.

```
"players": [  
  
  {  
  
    "name": "Ivan",  
  
    "balance": 100  
  
  },  
  
  {  
  
    "name": "Ivan",  
  
    "balance": 100  
  
  }  
  
],
```

3. Entity Object Specification

The game recognizes entities dynamically based on the exact string provided in entityType. The valid types are: "City", "Militia", "Infantry", "Cavalry", and "Artillery".

Base Parameters (Required for ALL entities)

- x / y: Integer coordinates. Must be ≥ 0 and strictly less than the map's width/height.
- ownerName: The string matching the player who controls this entity.
- id: String (Optional). If provided, it must be completely unique across the entire save file.

City Objects

Defined when entityType is exactly "City".

- cityLevel: String representing the upgrade state (e.g., "LEVEL_1"). Defaults to "LEVEL_1". Max level is "LEVEL_3"
- **Constraint:** Only **one** City can exist per (x, y) coordinate.

```
{
  "entityType": "City",
  "x": 1,
  "y": 1,
  "ownerName": "Ivan",
  "cityLevel": "LEVEL_1"
},
```

Troop Objects

Defined when entityType is "Militia", "Infantry", "Cavalry", or "Artillery".

- hp: Integer. Must be greater than 0 and cannot exceed the specific unit class's maximum health.
- hasMovedThisTurn: Boolean. Defaults to false.
- hasAttackedThisTurn: Boolean. Defaults to false.
- **Constraint:** Only **one** Troop can exist per (x, y) coordinate.

```
{
  "entityType": "Artillery",
  "x": 2,
  "y": 1,
  "ownerName": "Ivan",
```

```
"hp": 20,  
"hasMovedThisTurn": false,  
"hasAttackedThisTurn": false  
},
```

4. Critical Logical Constraints

If you are modifying this file manually, you must adhere to these rules:

1. **Referential Integrity:** An entity's ownerName must precisely match a defined player name in the players array.
2. **Anti-Stacking Rules:** A City and a Troop can occupy the exact same tile (x, y), but you cannot stack two Cities or two Troops on identical coordinates.
3. **Active Game Verification:** The game will not load a scenario that is already finished. Both players must own at least one "City". If a player has zero cities in the configuration, the engine considers them defeated and will abort the load process.